

of Questions: 30

Question #: 1

During childbirth, an obstetrician may elect to perform a pudendal nerve block to anesthetize the perineum and vagina. This can be done transvaginally by injecting the anesthetic near the pudendal nerve as it passes from the pelvis into the perineum. To locate the nerve, the physician inserts a finger into the vagina and palpates laterally to most likely locate which of the following landmarks?

- A. Ischial tuberosity
- B. Anterior superior iliac spine
- ✓C. Ischial spine
- D. Posterior fornix
- E. Posterior superior iliac spine

Rationale: The pudendal nerve and the internal pudendal vessels leave the pelvis below the piriformis muscle. They hook around the ischial spine to enter the perineum.

Question #: 2

A 27-year-old woman is brought to the emergency department by the paramedics because of 1-hour history of motor vehicle collision. Physical examination shows tender urinary bladder, loss of anal tone and perianal numbness. A urinary catheter is inserted to relieve urinary retention. Imaging studies show multiple fractures of pelvis and lower lumbar vertebrae. Which of the following nerves was most likely involved to produce urinary symptoms?

- A. Superior hypogastric

- ✓B. Pelvic splanchnic
- C. Pelvic splanchnic
- D. Lumbar splanchnic
- E. Pudendal

Rationale: The urinary bladder wall includes the detrusor muscle, and it receives both its motor and sensory innervation from parasympathetic nerve fibers transmitted by way of the pelvic splanchnic nerves from S2 to S4.

Question #: 3

A-1-week-old boy is brought to the physician by his mother for postnatal examination. Physical examination shows bilaterally testes are located in the inguinal canal. No other abnormalities noted. Which of the following structures is most likely facilitates the descent of the testes to the scrotal sac?

- A. Peritoneal evagination into the scrotal sac
- B. Remnant of the processus vaginalis
- C. Fascial extensions of the inguinal abdominal wall
- D. Tissue suspending the testes from the mesonephros
- ✓E. Cord extending between the testis and scrotum

Rationale: Each of the structures below are related to the descending the testes however only the gubernaculum facilitates the direction of the testes. In its absence the testes may remain anywhere along the path of descent. The gubernaculum (E correct) joins the lower pole of the testis to the scrotum. Option A describes the processus vaginal as the testes descends to begin its inguinal phase of descent. Option (B) is the tunica vaginalis which is the obliterated processus vaginalis. During the passage of the testes from the abdomen to the scrotum portions of the abdominal wall evaginate giving rise to the spermatic cords (C) which contains the vas deferens and the testicular nerves and vasculature. During development of the testes it eventually separates from the mesonephros with the interposing tissues forming the mesorchium (D).

Question #: 4

A 30-year-old woman comes the physician for a prenatal checkup. The patient has not used cigarettes or alcohol and plans to continue abstaining during pregnancy. Vital signs are within normal limits. Ultrasonography shows a deficient amount of amniotic fluid. Which of the following most likely contributed to this condition?

- A. Maternal diabetes
- B. Fetal meroencephaly
- C. Fetal anencephaly
- ✓D. Fetal renal agenesis

Rationale: Oligohydramnios is an inadequate amount of amniotic fluid. This condition is often associated with placental insufficiency and fetal urinary system abnormalities. Polyhydramnios is a condition of excess amniotic fluid, and is often associated with maternal diabetes, fetal meroencephaly, or fetal anencephaly

Question #: 5

A 35-year-old woman G2 P1 is 30 weeks pregnant, comes to the emergency department because of metrorrhagia (non-menstrual vaginal bleeding) for past 4 hours. Physical examination shows the patient to be normotensive with discolored conjunctiva. Doppler ultrasonographic examination shows a non-homogenous placenta protruding into the bladder. Which of the following most correctly describes the patient's condition?

- A. Placenta accreta
- B. Placenta increta
- ✓C. Placenta percreta
- D. Placenta previa

Rationale: In placenta accreta, there is abnormal adherence of placental chorionic villi to myometrium of the uterine wall. In placenta increta, placental chorionic villi invade deeply into the myometrium. In placenta percreta, the placenta penetrates the entire uterine wall through the perimetrium and is often associated with third trimester bleeding. Placenta previa is associated with late trimester bleeding but by itself involves the placenta covering the internal cervical os, not an abnormal depth

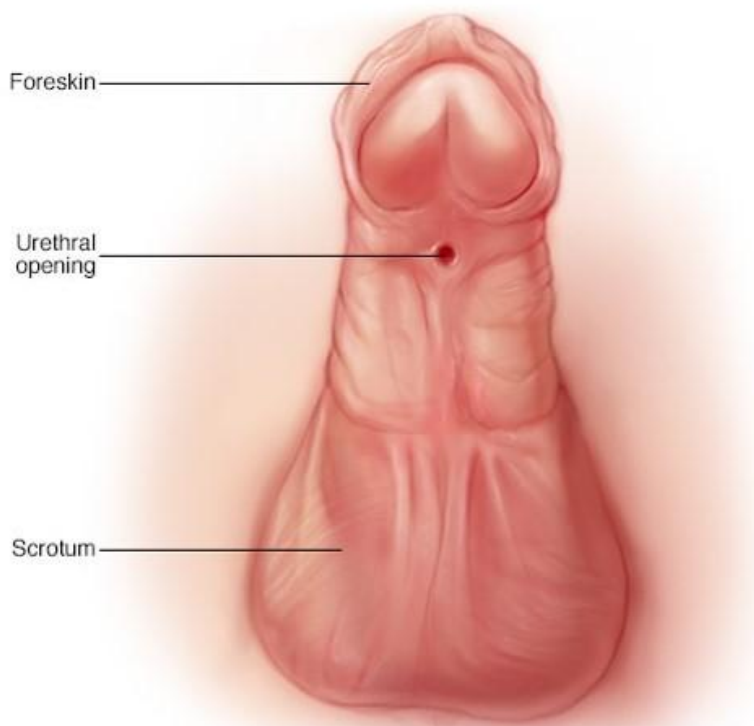
adherence.

Question #: 6

During a newborn checkup on a baby boy, a physician discovers the external urethral orifice is located in an ectopic location as shown in the attached diagram. Which of the following is the most likely diagnosis?

- A. Epispadias
- B. Cryptorchidism
- C. Perineal hypospadias
- ✓D. Penile hypospadias
- E. Hydrocele

Rationale: The correct answer is penile hypospadias, where the urethral opening is located on the ventral surface of the body of the penis as seen in the image. This is due to failure of fusion of the urethral folds. In perineal hypospadias, the urethral opening is located on the scrotum. This is due to failure of closure of the urethral and labioscrotal folds.



Question #: 7

A 36-year-old man comes to the emergency department because of a 12-hour history of perineal pain. He has difficulty urinating and says there is blood in his semen during ejaculation. Imaging studies show enlarged and inflamed seminal vesicles. Which of the following nerves transmits the visceral pain fibers from this organ back to the spinal cord?

- A. Lumbar splanchnics
- ✓B. Pelvic splanchnics
- C. Pudendal
- D. Ilioinguinal
- E. Obturator

Rationale: B: The pelvic pain line lies at the lower limit of the peritoneum in the pelvis. Visceral afferents from organs that lie below the pain line travel back to the spinal cord with parasympathetics. Visceral afferents from organs that lie above the pain line travel back to the spinal cord with sympathetics. The seminal vesicle lies below the pain line, so visceral afferents from this organ travels back to the spinal cord with pelvic splanchnics nerves.

Question #: 8

A 20-year-old woman comes to the physician because of painful, heavy bleeding during her last menstrual cycle. An ultrasound examination shows endometriomas (ectopic endometrial tissue) of the cervical canal and the uterus in the normal anteverted position. The angle between which of the following organs best describes this position?

- ✓A. Vagina and the cervix
- B. Uterine body and cervix
- C. Uterine body and bladder

- D. Cervix and the rectum
- E. Vagina and uterine body

Rationale: The uterus lies in an anteflexed, anteverted position. Anteverted refers to the angle between the cervix and the vagina, while anteflexed refers to the angle between the cervix and the body of the uterus.

Question #: 9

A 54-year-old man comes to the physician for a follow-up examination. He has a 2-year history of type 2 diabetes mellitus managed with metformin. He feels well and reports no current symptoms. He does not use tobacco, alcohol, or recreational drugs. Blood pressure is 120/74 mm Hg, pulse is 62/min, and BMI is 28 kg/m². Physical examination is unremarkable. Laboratory studies show hemoglobin A1c is 8.6% (Reference range: <6%). Because glycemic control remains suboptimal, the patient is prescribed pioglitazone, a hypoglycemic agent that enhances insulin sensitivity. Which of the following mechanisms best explains the desired effect of this medication?

- A. Decreased expression of GLUT-4
- B. Decreased expression of glucagon like peptide-1 (GLP-1)
- C. Increased hepatic glucose production
- ✓D. Increased production of adiponectin
- E. Increased production of interleukin-6

Rationale: Low adiponectin levels in patients with type 2 diabetes mellitus contribute to insulin resistance. The drug increases adiponectin release from adipocytes to improve insulin sensitivity.

When target tissues are insulin-sensitive:

GLUT-4 expression in muscle and adipocytes is high

GLP-1 (incretin) secretion is higher

Hepatic gluconeogenesis and glucose output are lower

Interleukin-6 (inflammatory mediator) levels are higher in obese patients with insulin resistance and contribute to the development of type 2 diabetes mellitus.

Question #: 10

A 38-year-old woman comes to the physician because of a 2-month history of progressive darkening of her skin. She reports generalized fatigue and unintentional weight loss. Blood pressure is 100/70 mm Hg. Physical examination shows diffuse hyperpigmentation of the skin, including the palmar creases and buccal mucosa. Laboratory studies show elevated plasma ACTH concentration and the presence of adrenocortical antibodies. Which of the following findings is most likely present in this patient?

- A. Increased renal sodium reabsorption
- B. Elevated serum sodium levels
- C. Increased renal calcium reabsorption
- ✓D. Decreased renal potassium excretion
- E. Elevated plasma glucose levels

Rationale: The vignette describes a patient with Addison's disease. This is characterized by autoantibodies destroying the adrenal cortex. Patients have low levels of glucocorticoids and mineralocorticoids. These patients generally have hypoglycemia.

Key findings in Addison's disease:

↓ Cortisol → loss of negative feedback → ↑ ACTH

↓ Aldosterone → hyponatremia, hyperkalemia, hypotension

↑ ACTH also leads to ↑ melanocyte-stimulating hormone (MSH) production (from POMC precursor) → diffuse hyperpigmentation

Low aldosterone secretion results in decreased sodium reabsorption from the renal tubule and increases the risk of hyponatremia and hypotension. Potassium secretion into the renal tubule decreases, leading to high serum K⁺ levels.

Hyperparathyroidism results in hypercalcemia due to increased renal calcium reabsorption.

Vitamin D toxicity can also result in hypercalcemia due to increased dietary calcium absorption.

Excessive aldosterone secretion (Primary hyperaldosteronism) activates the H⁺-secretion in the renal tubule and results in metabolic alkalosis.

Cushing syndrome (Increased glucocorticoid secretion) presents with hyperglycemia.

Question #: 11

A 2-week-old infant is brought to the emergency department because of a 6-hour history of poor feeding and lethargy. The parents report that the infant has had vomiting and decreased urine output over the past day. Blood pressure is 70/40 mm Hg. The infant appears severely dehydrated. Physical examination shows ambiguous genitalia. Laboratory studies are shown in the table. Which of the following enzyme deficiencies is most likely responsible for this patient's condition?

- A. 3-beta hydroxysteroid dehydrogenase
- B. 17-hydroxylase (CYP 17)
- ✓C. 21-hydroxylase (CYP 21)
- D. 11-hydroxylase (CYP 11B)
- E. Aromatase (CYP 19)

Rationale: The vignette describes a patient with congenital adrenal hyperplasia due to a deficiency in 21-hydroxylase. Note the hypotension, low serum Na⁺ levels, increased serum K⁺ levels, and increased plasma renin activity. These features are due to aldosterone deficiency; Cortisol deficiency results in hypoglycemia. Also, please remember the elevated levels of 17-hydroxyprogesterone (a substrate of CYP 21 (21 hydroxylase)).

Other enzyme defects:

11 β -hydroxylase deficiency: $\uparrow$ 11-deoxycorticosterone (mineralocorticoid effect) $\rightarrow$ hypertension, not salt wasting.

17 α -hydroxylase deficiency: $\downarrow$ sex steroids $\rightarrow$ sexual infantilism, hypertension, hypokalemia.

3 β -hydroxysteroid dehydrogenase deficiency affects both glucocorticoid and sex steroid synthesis; genital ambiguity in both sexes, but without an elevated 17-hydroxyprogesterone pattern.

Aromatase deficiency affects females $\rightarrow$ Excessive androgens $\rightarrow$ Ambiguous genitalia in infants; failure to develop secondary sex characteristics and hirsutism as the children grow older.

Laboratory Test	Result	Reference Range
Serum sodium	126 mEq/L	136-146 mEq/L
Serum potassium	5.8 mEq/L	3.5-5.0 mEq/L
Serum glucose	45 mg/dL	70-100 mg/dL
Plasma renin activity	Increased	—
Serum cortisol	Decreased	—
Serum 17-hydroxyprogesterone	Increased	—

Question #: 12

A 50-year-old man comes to the physician because of a 2-month history of intermittent episodes of lightheadedness and fainting. The episodes typically occur in the morning and improve after drinking a glass of orange juice. BMI is 21 kg/m², and the patient has lost 4.5 kg (10 lb) since the previous visit 4 months ago. Laboratory studies show fasting serum glucose concentration is 60 mg/dL (reference range: 70–100 mg/dL). Abdominal imaging shows a mass involving specific cells of the pancreatic islets of Langerhans. Additional laboratory studies show elevated concentrations of C-peptide and a specific peptide hormone. Which of the following best describes the signal transduction mechanism of this hormone?

- A. Activates protein kinase A
- B. Increases intracellular calcium levels
- C. Activates cGMP-dependent protein kinase
- D. Binds to intracellular receptors
- ✓E. Activates receptor tyrosine kinase

Rationale: The vignette describes a patient with fasting hypoglycemia due to increased hormone secretion from the islets of Langerhans. Note the elevated C-peptide levels. The patient most likely has an insulinoma. Insulin binds to its receptor on the plasma membrane, which activates the tyrosine kinase activity in the intracellular domains of this receptor. This, in turn, results in phosphorylation of insulin receptor substrates-1 (IRS-1) and activation of PI3-Kinase.

Glucagon binds to its receptors on the cell membrane to activate the Gs-alpha subunit and activate adenylate cyclase. Adenylate cyclase increases intracellular cAMP levels, activating Protein kinase A.

Epinephrine and norepinephrine bind to alpha-1 adrenergic receptors, activating the Gq-alpha subunit, activating phospholipase C, and increasing intracellular IP3 and DAG levels. IP3, in turn, increases intracellular calcium levels.

Nitric oxide activates soluble guanylate cyclase and increases the intracellular cGMP levels, activating cGMP-dependent protein kinase (Protein kinase G).

Question #: 13

A 45-year-old man comes to the physician because of a 6-month history of unintentional weight loss despite an increased appetite. He has lost 4.5 kg (10 lb) since his previous visit 6 months ago. His BMI is 22 kg/m². Laboratory studies show fasting serum glucose concentration is 200 mg/dL (reference range: 70-100 mg/dL). Abdominal imaging shows a mass involving specific cells of the pancreatic islets of Langerhans. Activation of which of the following intracellular signaling proteins is most likely mediated by the hormone secreted by this tumor?

- ✓A. Adenylate cyclase
- B. Protein kinase C
- C. cGMP-dependent protein kinase
- D. Phosphodiesterase
- E. Tyrosine kinase
- F. Guanylate cyclase

Rationale: The patient in the vignette has hyperglycemia due to a tumor from the alpha cells of the pancreas (glucagonoma). Glucagon signal transduction is via the activation of adenylate cyclase, which increases the intracellular cAMP (second messenger). cAMP activates protein kinase A.

Tyrosine kinase is activated by insulin. However, remember, patients with insulinoma have hypoglycemia (and not hyperglycemia).

Question #: 14

A 12-year-old girl is brought to the physician by her mother because of a 1-month history of weight loss, fatigue, excessive thirst, and urination. Her mother also reports burning during urination. There is no family history of major medical illnesses. Physical examination shows lethargy, dry oral mucosa, and decreased skin turgor. Which of the following laboratory findings is most likely in this patient?

- A. Plasma glucose of 100 mg/dl
- B. Urine glucose absent
- ✓C. Plasma glucose of 250 mg/dl
- D. Plasma glucose of 40 mg/dl
- E. HbA1c of 5%

Rationale: The vignette describes a patient with type 1 diabetes mellitus. Note the weight loss, polyuria, polydipsia, and dehydration. Hyperglycemia is a cardinal diagnostic feature of both types of diabetes mellitus. The elevated plasma glucose exceeds the renal threshold for glucose and hence glucose is found in the urine (glucosuria). HbA1c (glycated hemoglobin) levels are elevated in type 1 and type 2 diabetes mellitus. Ketone body levels would also likely be elevated in blood and urine (Ketonemia and ketonuria).

Hypoglycemia is characterized by palpitations, sweating, tremors, and seizures if the plasma glucose levels are too low.

Question #: 15

A 32-year-old man comes to the physician because of 2-week history of pain and swelling in his left scrotum. A biopsy specimen from the left scrotal swelling is shown in the attached photomicrograph. Which of the following functions is most likely affected due to damage of this structure?

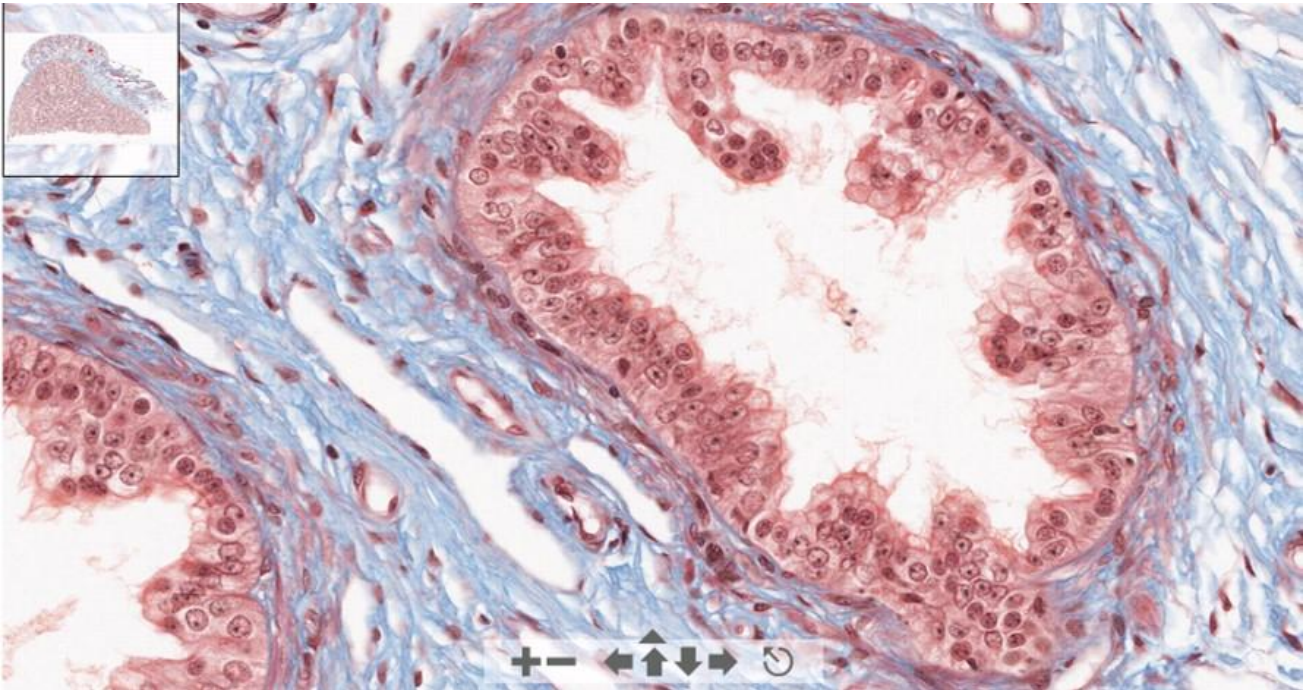
- A. Blood-testis barrier
- ✓B. Reabsorption of seminiferous fluid

- C. Testosterone production
- D. Sperm storage
- E. Reduced semen volume production

Rationale: The efferent ductule is lined by pseudostratified columnar epithelium and surrounded by a thin smooth muscle layer. The epithelium also consists of short cells with abundant microvilli on their surface, that are responsible for the reabsorption of most of the fluid secreted in the seminiferous tubules. The columnar cells have cilia on their surface. Interspersed between the columnar and short cells are basal cells which act as stem cells for the renewal of the other epithelial cells.

This arrangement of tall columnar and short cells give the lumen a "saw-toothed" appearance, which can be appreciated in the attachment.

The blood-testis barrier is formed by tight junctions between neighboring Sertoli cells. Sertoli cells are found in the seminiferous epithelium, which is not depicted in the micrograph. The Leydig cells are responsible for testosterone production, and is found in the interstitium between lobules in the testis. The epididymis does sperm storage. If the seminal vesicle is affected, then that would affect semen volume.



Question #: 16

A researcher is investigating the effects of a novel compound on cells of the seminiferous epithelium. He discovers that the compound affects

spermatogenesis, particularly cells that are in their first phase of meiotic division. Which of the following cells are most likely affected?

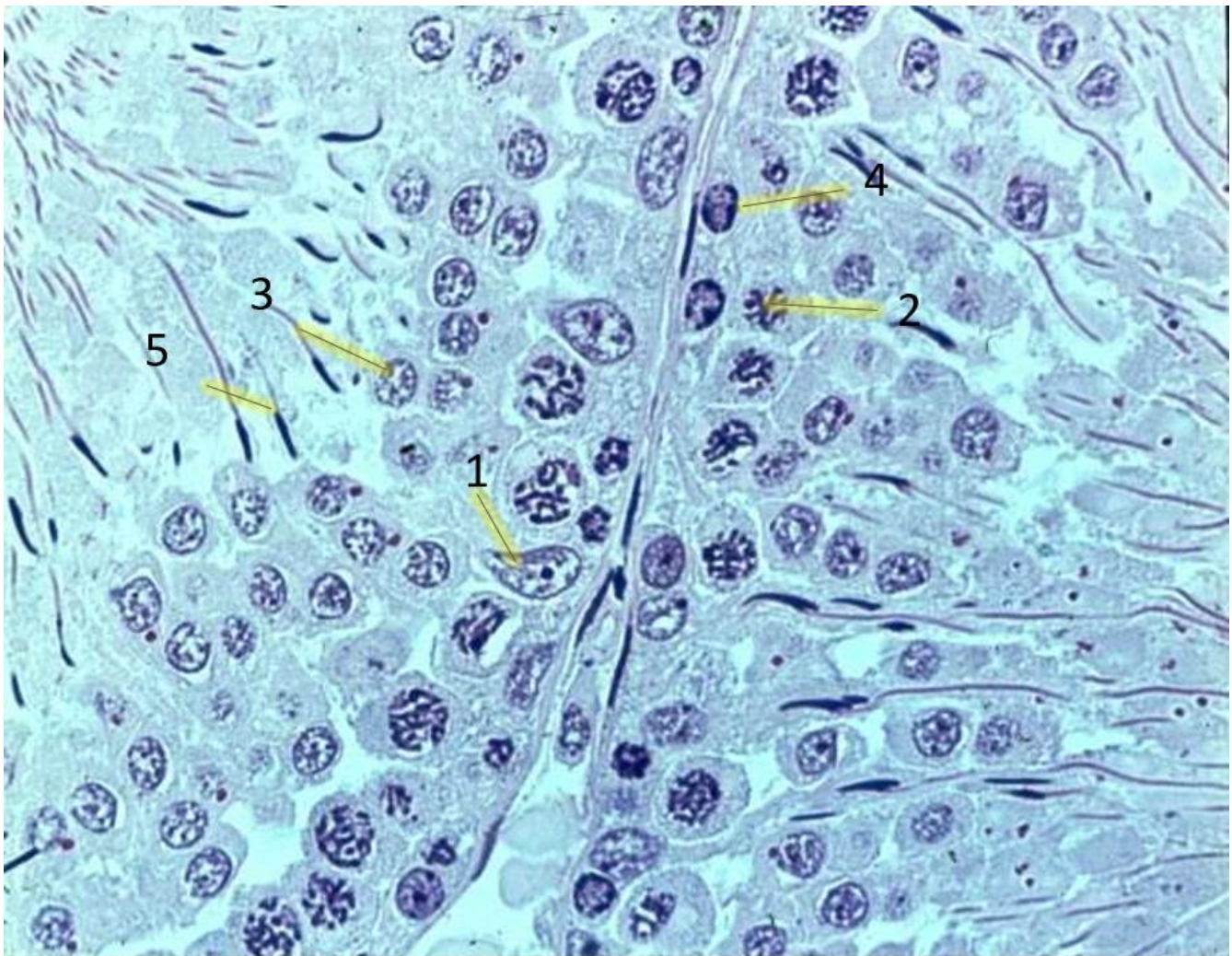
- A. 1
- ✓B. 2
- C. 3
- D. 4
- E. 5

Rationale: 1-nucleus of Sertoli cells

4- Spermatogonia - these divide by mitotic activity to replicate themselves and to produce primary spermatocytes (2).

Primary spermatocytes (2) enter meiosis I and form secondary spermatocytes that go into meiosis II giving rise to haploid spermatids (3).

5 - spermatozoa



Question #: 17

A 25-year-old man comes to the physician with his wife because of a 1-year history of unsuccessful conception. Physical examination shows no abnormalities. Laboratory studies shows reduced levels of testosterone. Which of the following hormones most likely stimulates the production of the affected hormone in this patient?

- A. Estrogen
- B. Inhibin
- C. Progesterone
- D. Follicular stimulating hormone
- ✓E. Luteinizing hormone

Rationale: Leydig cells are interstitial cells responsible for the secretion of the steroid hormone, testosterone. Leydig cells are found in the stroma of the testes. They are able to produce testosterone in the presence of luteinizing hormone, which binds to the LH receptors on the cell.

Question #: 18

A research group is investigating the effect of a synthetic form of FSH on spermatogenesis and cells of the seminiferous tubules in mice. The drug is administered over a period of two to three months to the subjects following which a marked increase in spermatozoa is observed in the semen. Biopsy samples of testicular tissue is provided in the attached photomicrograph. Which of the following cells is the most likely target of this drug?

- A. 1
- B. 2
- C. 3
- ✓D. 4
- E. 5

Rationale: Sertoli cells have a pear shaped nucleus and is found throughout the entire length of the seminiferous epithelium. These cells have receptors for FSH and secrete inhibin under FSH stimulation.

1 - primary spermatocyte (coarse clumps of chromatin within the cytoplasm)

2 - type A pale - based on location and lightly staining chromatin within the cytoplasm

3 - interstitial cells of Leydig

5- late spermatid



Question #: 19

A 65-year-old woman comes to the physician because of a 3-month history of bright red blood per vagina and a 6-month history of unintentional weight loss of 13.6-kg (30-lb). She has been menopausal for the past 10 years. An ultrasound scan shows an abnormally thick endometrium. Microscopic examination of a biopsy specimen of the endometrium shows metaplastic changes to the epithelial cells. Which

of the following classifications best describes the epithelium in which the metaplastic changes were observed?

- A. Transitional
- B. Simple squamous
- ✓C. Simple columnar
- D. Simple cuboidal
- E. Stratified squamous

Rationale: The endometrium of the uterus is made up of simple columnar cells.

Transitional epithelium is unique to certain parts of the urinary system. Simple squamous epithelia lines sites such as blood vessels and serous cavities. Simple cuboidal epithelium can be found in the thyroid gland, surrounding thyroid follicles. Stratified epithelium will be found in regions that need more protection. This type of epithelium facilitates more absorption and secretion more than protection. Stratified squamous can be found in the vagina in females.

Question #: 20

A 28-year-old woman who has been diagnosed with leukemia decides to have her oocytes harvested and frozen prior to chemotherapy. Prior to harvesting she is given medications to stimulate follicular growth. Subsequently, oocytes were harvested from pre-ovulatory follicles. The harvested oocytes **just prior** to ovulation will be in which of the following stages of cell division?

- A. Anaphase of 2nd meiotic division
- B. Prophase of 1st meiotic division
- C. Prophase of 2nd meiotic division
- D. Anaphase of 1st meiotic division
- ✓E. Metaphase of the 2nd meiotic division

Rationale: The primary oocyte is arrested in prophase of meiosis I. This process is carried out in the embryo. The oocyte remains in prophase of meiosis I until just before ovulation when it completes this division, and enters meiosis II. It becomes stuck in metaphase of meiosis II. This is not completed except there is fertilization.

Question #: 21

A 30-year-old woman comes to the physician for a scheduled radical hysterectomy with oophorectomy. She has a history of metastatic cancer diagnosed 6-weeks ago. Which of the following hormones is secreted by the cells indicated by the arrow?

- A. Progesterone
- ✓B. Estrogen
- C. Luteinizing hormone
- D. Androstenedione
- E. Follicle stimulating hormone

Rationale: Under the influence of FSH, granulosa cells (indicted by the arrow) convert androgens (androstenedione) (produced by theca interna) to estrogen. Both LH and FSH are secreted by cells of the pituitary gland. The corpus luteum secretes progesterone.



Question #: 22

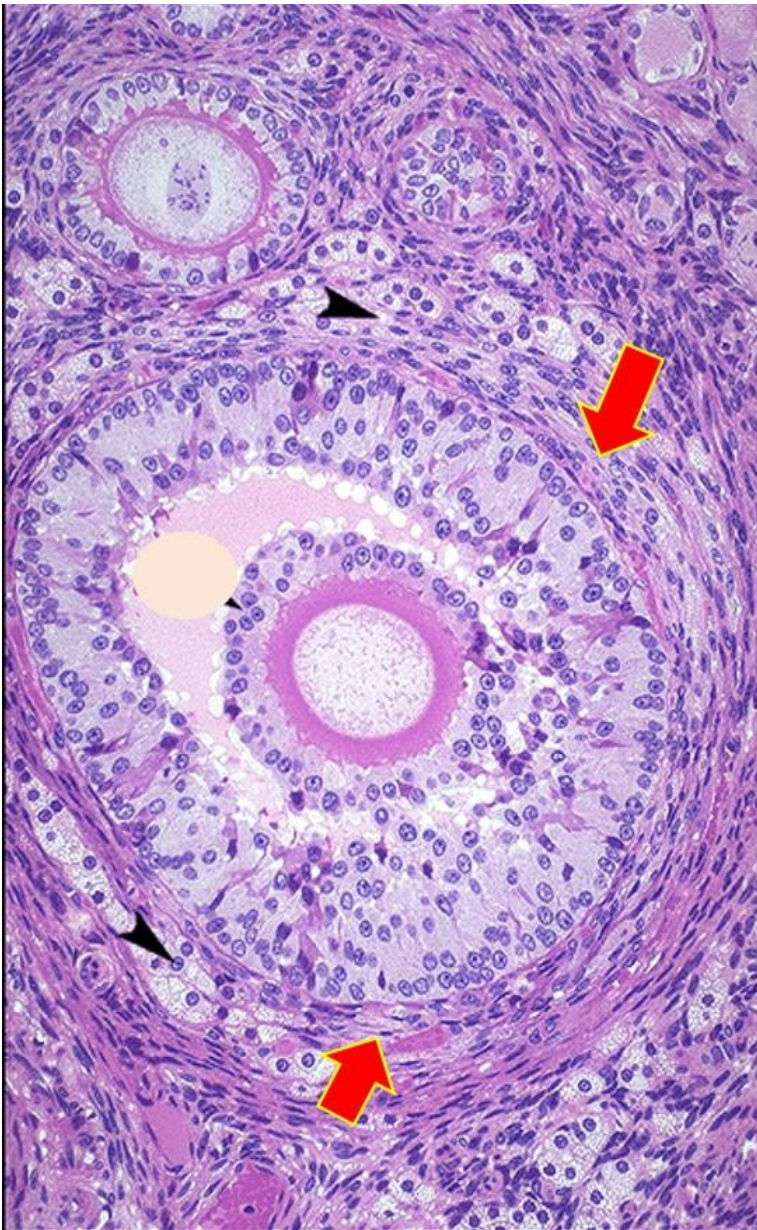
A 45-year-old woman comes to the physician because of a 4-month history of persistent pelvic pain, worse on sexual intercourse and irregular menstrual cycles. Pelvic CT-scan shows a suspicious mass on the right ovary. Microscopic examination of the ovarian mass shows abnormal cellular structures. A normal photomicrograph is attached and the cells most affected are highlighted by red arrows. Which of the following best identifies a function of these highlighted cells?

- A. Develop and mature into an egg
- B. Protection of the developing oocyte
- C. Formation of the corpus luteum
- ✓D. Synthesize and secrete androstenedione
- E. Production of estrogen hormone

Rationale: The red arrows are pointing at the theca interna cells of a secondary follicle. Theca interna cells are steroid-producing cells and possess a large number of LH receptors. They also possess receptors for LDL, which allows them to pick up LDL from the blood. Cholesterol is liberated from the LDL, and used by the theca interna cells to produce the androgen androstenedione. This enzyme serves as the precursor to estrogen.

The granulosa cells, under the influence of FSH, pick up the androstenedione secreted from the theca interna cells, and in the presence of the enzyme aromatase, convert it into estradiol/estrone/estrogen (option E). The zona pellucida protects the developing oocyte. The corpus luteum is formed from the mature follicle, after ovulation, and not from the theca interna cells. Option A represents the function of the oocyte.

A 45-year-old female presents with persistent pelvic pain and irregular menstrual cycles. Imaging reveals a suspicious mass on the right ovary, prompting a diagnostic laparoscopic surgery and biopsy. Microscopic examination of the ovarian tissue reveals abnormal cellular structures, and the pathologist notes the presence of cells highlighted by red arrows. The pathologist's report highlights these cells as having a specific function.



Question #: 23

A 42-year-old man comes to the physician because of a 4-month history of headache, 6.8-kg (15-lb) weight gain, round face, and excess fat in the back of the neck. His blood pressure is 162/92 mm Hg; other vital signs are within normal limits. Physical examination shows truncal and facial obesity, thin extremities, abdominal striae, and hyperpigmentation of skin. Laboratory studies show elevated blood glucose concentration. MRI of brain is pending. Which of the following serum hormonal

alterations is most likely in this patient?

Option	Cortisol	Adrenocorticotropic hormone (ACTH)	Corticotropin releasing hormone (CRH)
A	Increase	No change	Increase
B	Increase	Decrease	No change
C	Decrease	Increase	Increase
D	Decrease	Increase	Decrease
E	Increase	Increase	Decrease

- A. A
- B. B
- C. C
- D. D
- ✓E. E

Rationale: This patient has elevated ACTH that increases secretion of cortisol and suppresses secretion of hypothalamic CRH. Additionally, an elevated plasma cortisol also suppresses hypothalamic CRH secretion. Same elevated plasma cortisol is not able to suppress anterior pituitary because the anterior pituitary with ACTH-secreting tumor does not respond physiologically. This patient has features of Cushing's syndrome along with hyperpigmentation. This is enough to diagnose elevated ACTH. The source of ACTH may be pituitary or an ectopic site.

Question #: 24

A 32-year-old man is brought to the emergency department 1 hour after he was involved in a motor vehicle collision in which he sustained multiple fractures to the right upper and lower extremities and head. He was traveling at 70 mph when he lost control and hit a concrete wall. He is unconscious and bleeding from his wounds. His temperature is 37.0°C (98.6°F), pulse is 116/min, respirations are 18/min, and blood pressure is 114/72 mm Hg. Physical examination shows abrasions over the face, neck, right shoulder, and thigh. Cranial nerves are intact. He is

admitted to the intensive care and monitored. Twelve hours later, his condition deteriorates, and he develops traumatic brain injury. He is under severe stress. Which of the following sets of hormonal alterations is most likely in this patient?

Option	Cortisol	Adrenocorticotropic hormone (ACTH)	Corticotropin releasing hormone (CRH)
A	Increased	Increased	Decreased
B	Increased	Increased	Increased
C	Decreased	Increased	Decreased
D	Decreased	Decreased	Increased
E	Increased	Decreased	Decreased
F	Decreased	Decreased	Decreased

- A. A
- ✓B. B
- C. C
- D. D
- E. E
- F. F

Rationale: This patient is in a traumatic stress condition due to head injury and physical trauma. Under stress condition, the hypothalamo-pituitary-adrenal axis is stimulated. This results in increased secretion of all hormones in this axis.

Question #: 25

A 21-year-old woman comes to the physician because of a 3-month history of weight loss despite an increased appetite, sweaty palms, and prominent eyes. Her blood pressure is 140/76 mm Hg, respirations are 18/min, and pulse is 106/min. Physical examination shows a diffuse goiter. A thyroid scan shows increased radio-iodine uptake. She is diagnosed with Graves' disease. Which of the following is most likely

expected in this patient?

- A. Decreased coupling of monoiodotyrosine (MIT) and diiodotyrosine (DIT)
- B. Decreased thyroid hormone binding globulins
- C. Increased thyroid stimulating hormone (TSH) and thyrotropin releasing hormone (TRH) levels
- ✓D. Stimulation of the Na/I symporter
- E. Decreased synthesis of thyroglobulin

Rationale: In Grave's disease she has TSH receptor stimulating antibodies so all steps in TSH synthesis are enhanced, and not decreased. However her TSH & TRH will be suppressed due to negative feedback effect from high levels of T3 & T4.

Question #: 26

A 48-year-old man comes to the physician because of a 6-month history of increasing shoe size and hat size, prominent growth of the nose and thickening of fingers. His blood pressure is 146/91 mm Hg and pulse is 78/min. Physical examination shows a prominent forehead, large nose, protruding mandible, enlarged hands and feet, and coarse skin. Laboratory studies show elevated fasting blood glucose and plasma insulin-like growth factor-1. Under physiologic conditions, which of the following hormones is most likely released in response to an increased plasma IGF-1?

- A. Growth Hormone
- B. Growth Hormone Releasing Hormone
- ✓C. Somatostatin
- D. Prolactin
- E. Ghrelin

Rationale: IGF-1 feeds back positively on to the hypothalamus to increase the release of somatostatin. It suppresses the GHRH and GH, and has no effects on the other hormones listed in the answer options.

Question #: 27

A 36-year-old man comes to the physician because of a 5-week history of progressive enlargement of his left scrotum. There is no history of infection. His vital signs are within normal limits. Physical examination shows a hard mass in the left scrotum. Testicular cancer is suspected and both testes are surgically removed. Histologic examination of a biopsy specimen of this mass confirms seminoma testis. A decrease in which of the following hormones is most likely in this patient following the surgery?

- A. Gonadotropin releasing hormone (GnRH)
- B. Follicular stimulating hormone (FSH)
- ✓C. Inhibin
- D. Adrenal androgens
- E. Luteinizing hormone

Rationale: Removal of the testes, results in a loss of Leydig and Sertoli cells. The Leydig cells secrete testosterone and the Sertoli cells secrete inhibin.

Question #: 28

A 22-year-old woman comes to the physician because of a 2-month history of heart palpitations, sweaty palms, rapid weight loss, eyes appearing bigger than usual and feeling anxious and fidgety. Her blood pressure is 136/80 mmHg, pulse is 100/min, and respirations are 14/min. Physical examination shows protruding eyes with a staring appearance. She is suspected of having Graves' disease. The elevated hormone responsible for her symptoms would trigger a response in its target cell by which of the following most likely mechanisms?

- A. IP3 and DAG second messenger system
- B. cAMP second messenger system

- C. Cell depolarization
- ✓D. Nuclear receptor binding
- E. Tyrosine kinase activation

Rationale: TSH stimulates the synthesis of thyroid hormone. Thyroid hormone binds to the nuclear receptors and promotes DNA transcription to have thyroid hormone like effects.

Question #: 29

A 23-year-old man comes to the physician because of a 4-month history of erectile dysfunction. He takes no medications. Physical examination shows no abnormalities. Laboratory studies show normal levels of testosterone, LH and FSH. The physician diagnoses that his erectile dysfunction is related to performance anxiety. He is treated with psychotherapy which normalizes his erectile function. The physiological pathway for penile erection most likely depends on which of the following mechanisms?

- A. Increased inhibin formation in penile tissue
- B. Stimulation of calcium release in penile smooth muscle
- C. Inhibition of the cGMP in penile smooth muscle
- ✓D. Stimulation of the nitric oxide (NO) formation in penile tissue
- E. Stimulation of cAMP in penile smooth muscle

Rationale: Penile erection is due to vasodilation via nitric oxide (NO) and/or Ach. Efficacy of psychotherapy on erectile dysfunction (ED) would necessitate an increase in NO.

Ejaculation would only occur after the ED is resolved

Urination or micturition would not solve the ED

cGMP results in relaxation of smooth muscle, and inhibition would prevent vasodilation and erection in the penis.

Question #: 30

A 34-year-old woman comes to the physician because of a 6-month

history of passing large volume of urine with increased frequency of urination and increased thirst. She was diagnosed with sarcoidosis 4 months ago and is on treatment. Her pulse is 98/min, blood pressure is 112/72 mm Hg, and respirations are 16/min. Physical examination shows increased skin turgor and dry lips and tongue. Laboratory studies show:

Test	Patient value (Reference value)
Serum sodium	154 mEq/L (Normal: 136-146 mEq/L)
Serum calcium	12.8 mg/dL (Normal: 8.5-10.5 mg/dL)
Random blood glucose	90 mg/dL (Normal: 80-120 mg/dL)
Serum osmolality	305 mOsmol/kg H ₂ O (Normal: 275-295 mOsmol/kg H ₂ O)
24-hour urine volume	5800 mL/day (Normal: 800-2000 mL/day)
Urine osmolality	105 mOsmol/kg H ₂ O (Normal: 50-1200 mOsmol/kg H ₂ O)
Urine specific gravity	1.010 (Normal: 1.005-1.030)

She is admitted and a water deprivation test is performed on her. She is restricted from drinking fluids for 2 hours and is administered an injection of desmopressin (synthetic vasopressin). Urine osmolality measured 30 min and 60 min after this injection shows osmolality of 106 mOsmol/kg H₂O and 105 mOsmol/kg H₂O respectively. Which of the following is the most likely diagnosis in this patient?

- A. Psychogenic polydipsia
- B. Syndrome of inappropriate ADH (SIADH)
- ✓C. Nephrogenic diabetes insipidus
- D. Cerebral salt wasting syndrome
- E. Neurogenic diabetes insipidus

Rationale: Clinical features and lab show that she has diabetes insipidus (DI). To find out what type of DI (neurogenic or nephrogenic), a water deprivation test with injection of a synthetic vasopressin is done. In neurogenic DI, vasopressin synthesis is affected and therefore, injecting vasopressin would increase the osmolality of the urine. In nephrogenic DI, the response of kidney to vasopressin is

defective, and therefore, injection of vasopressin does not increase urine osmolality. This is what happening in this patient.